

Chapter 18: No-Cloning Theorem and Quantum Teleportation I

Kamonluk Suksen Ph.D.

No-Cloning Theorem

- **No-Cloning Theorem** is a very important property of quantum mechanics.
- It says that not all quantum states can be cloned or copied.
- We can copy the contents of any classical register or memory to another classical register or memory. But quantum mechanics tells us that we cannot copy any arbitrary quantum state from a system to another identical system.
- In the electronic age, while we can copy the information, we are not copying the matters. The memory in our computer is not exactly the same as the memory in our cell phone, at least at the molecular level. So, we are not really cloning a system when we copy a file from our cell phone to our computer. We only copy the information.
- Copying a state in the microscopic scale, where quantum mechanics dominates has more meaning than just copying information on the macroscopic scale.
- Assume there are two electrons, A and B . Their initial states are $|\psi\rangle_A$ and $|0\rangle_B$. We do not know what $|\psi\rangle_A$ is but we know it must be a linear combination of $|0\rangle$ and $|1\rangle$. That is

$$|\psi\rangle_A = \alpha|0\rangle_A + \beta|1\rangle_A$$

No-Cloning Theorem

- The state of the whole system (i.e. electrons A and B) is a tensor product of both, which is $|\psi\rangle_A \otimes |0\rangle_B = |\psi\rangle_A |0\rangle_B$. Can we clone the state (wavefunction) from electron A to electron B ? If so, it means that after cloning, the state of the whole system becomes $|\psi\rangle_A |\psi\rangle_B$.
- In a classical computer, we need to read before copying. But in a quantum computer, we cannot clone by reading the state first because the state will collapse to either $|0\rangle$ or $|1\rangle$, after which we would have destroyed the original state and thus it is not a cloning operation.
- Cloning must be done through an operation (not a measurement) and we can write it as an operator (matrix) like other quantum gates, and it must be unitary (if it exists). It must also be a rotation of the vector in the corresponding hyperspace.
- Let the cloning operator be L , which is an operator in the $\mathbb{C}^4$ space. We have

$$L(|\psi\rangle_A |0\rangle_B) = |\psi\rangle_A |\psi\rangle_B$$

We can expand the final state as

$$\begin{aligned} L(|\psi\rangle_A |0\rangle_B) &= |\psi\rangle_A |\psi\rangle_B = (\alpha|0\rangle_A + \beta|1\rangle_A) \otimes (\alpha|0\rangle_B + \beta|1\rangle_B) \\ &= (\alpha|0\rangle_A \alpha|0\rangle_B + \alpha|0\rangle_A \beta|1\rangle_B + \beta|1\rangle_A \alpha|0\rangle_B + \beta|1\rangle_A \beta|1\rangle_B) \\ &= \alpha^2 |0\rangle_A |0\rangle_B + \alpha\beta |0\rangle_A |1\rangle_B + \beta\alpha |1\rangle_A |0\rangle_B + \beta^2 |1\rangle_A |1\rangle_B \\ &= \alpha^2 |00\rangle + \alpha\beta |01\rangle + \beta\alpha |10\rangle + \beta^2 |11\rangle \end{aligned}$$

No-Cloning Theorem

- On the other hand, the initial state before cloning can also be written as

$$\begin{aligned} |\psi\rangle_A |0\rangle_B &= (\alpha|0\rangle_A + \beta|1\rangle_A)|0\rangle_B = \alpha|0\rangle_A|0\rangle_B + \beta|1\rangle_A|0\rangle_B \\ &= \alpha|00\rangle + \beta|10\rangle \end{aligned}$$

Since the cloning operator is just a regular operator, it obeys the rules of matrix vector multiplication including the linearity and distribution law. Therefore, after cloning on the state in the above equation, we have the following:

$$\begin{aligned} L(\alpha|00\rangle + \beta|10\rangle) &= L\alpha|00\rangle + L\beta|10\rangle \\ &= \alpha|00\rangle + \beta|11\rangle \\ &= \alpha|00\rangle + 0|01\rangle + 0|10\rangle + \beta|11\rangle \end{aligned}$$

We can see that $L(|\psi\rangle_A |0\rangle_B)$ is not equal to $L(\alpha|00\rangle + \beta|10\rangle)$.

Comparing the coefficients of $|01\rangle$, it means $\alpha\beta = 0$, which further means either $\alpha = 0$ or $\beta = 0$. This means that we can only clone the basis vectors $|1\rangle$ ($\alpha = 0$) or $|0\rangle$ ($\beta = 0$) but not any general vector.

- If electron A is in a basis state ($|0\rangle$ or $|1\rangle$), we can read it without destroying its state (because the basis state will stay the same after any measurement) and then just use any appropriate method to set electron B to that basis state to complete the cloning process.
- So, it is possible to clone the basis states but **NOT** other superposition states.

No-Cloning Theorem

- The no-cloning theorem has many important applications in quantum computing and quantum communication.
- For example, an eavesdropper will not be able to copy your message without destroying your original information. We can thus use this to avoid eavesdropping in quantum communication.
- The no-cloning theorem also prevents the design of any quantum algorithms that will allow useful information to propagate faster than the speed of the light.

Quantum Teleportation

- What is teleportation?
- It is the transfer of matters and energies without traversing the space in between.
- This also leads us to associate the transfer of matters and energies faster than the speed of light. We know this is contradicting the Theory of Relativity.
- Assume Alice has an electron with the state $|\psi\rangle_A = \alpha|0\rangle_A + \beta|1\rangle_A$. This is a general state that no one knows what α and β are.
- The purpose of quantum teleportation is to transfer the state to Bob's electron, which might have an initial state of $|0\rangle_B$. Here we use subscripts A and B to make it clear that the states belong to the electrons of Alice and Bob, respectively.
- We want to transfer the state of Alice's electron to Bob in no time regardless of the distance between them!
- However, do not forget the **no-cloning theorem**. When the state of Alice's electron is transferred to Bob's electron, the original state in Alice's electron will be changed (i.e. no longer $|\psi\rangle_A$). Otherwise, we violate the no-cloning theorem.
- Similarly, if you are quantum teleported to Mars, the 1000 particles on the Earth will lose their original states and you are no longer who you are on the Earth.

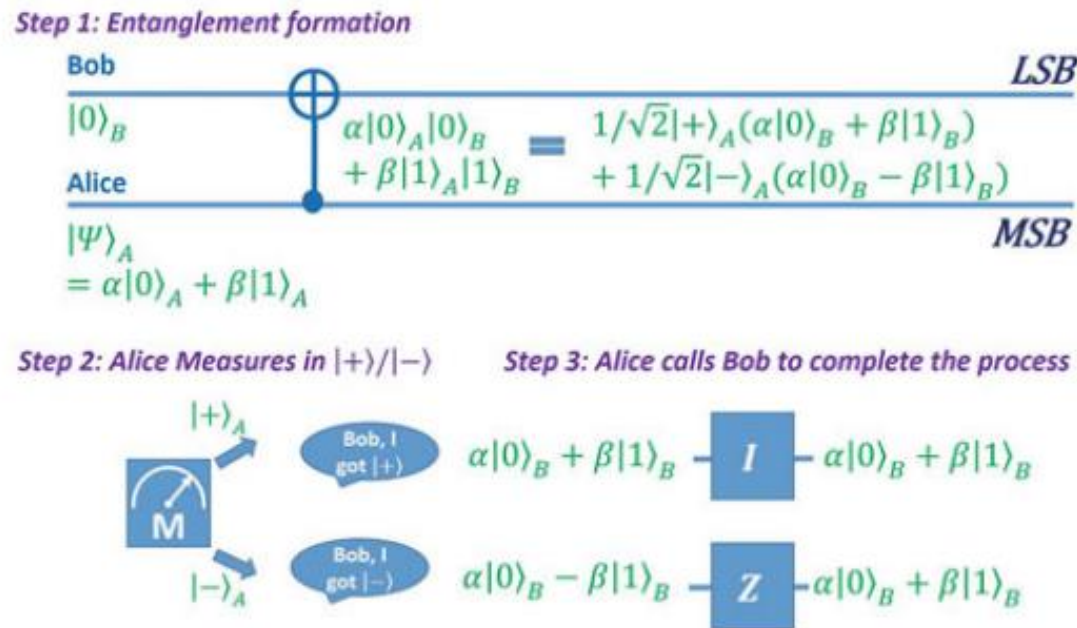
Quantum Teleportation

- In the quantum teleportation process, it seems that the state information will be teleported in no time. However, the information can only be confirmed (i.e. completion of the information transfer process) through two classical communications.
- Since classical communication cannot be faster than the speed of light, the whole quantum teleportation process is only completed at a speed slower than light. Therefore, it does not violate the Theory of Relativity.
- But we still call it teleportation because the action of copying the quantum state is instantaneous regardless of the distance.
- In the simplified version, only one classical communication is needed.

Quantum Teleportation

The algorithm for a simplified version of quantum teleportation:

- Step 1: Entanglement Between $|\psi\rangle_A$ and $|0\rangle_B$, we can use a CNOT gate to achieve this purpose.



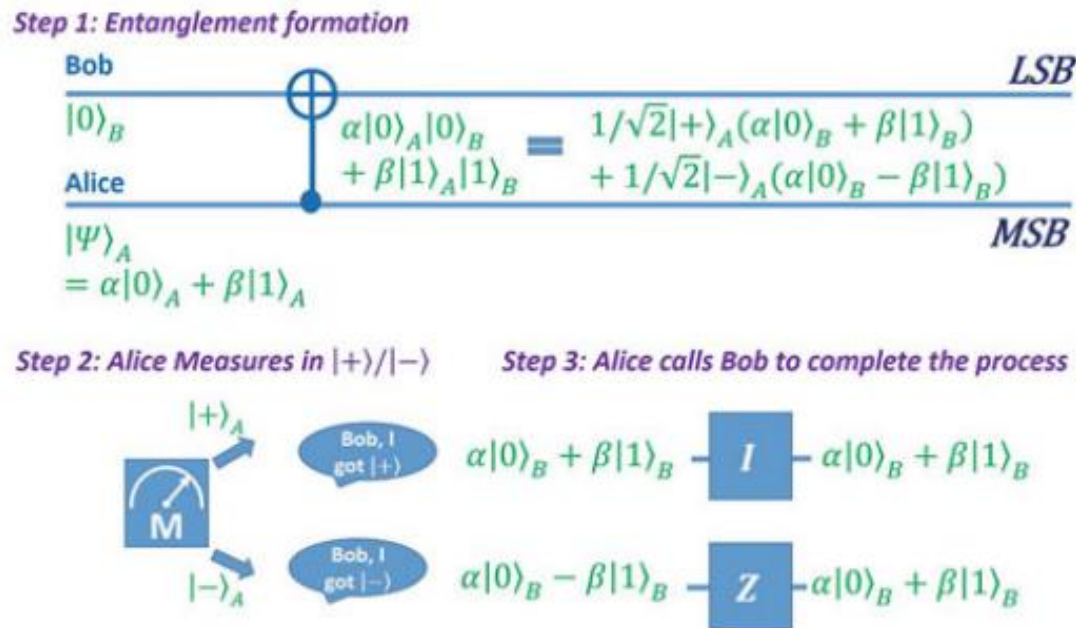
1. We start from Alice's electron and Bob's electron form a $\mathbb{C}^4$ space and their state is $|\psi\rangle_A \otimes |0\rangle_B = \alpha|0\rangle_A|0\rangle_B + \beta|1\rangle_A|0\rangle_B = \alpha|00\rangle + \beta|11\rangle$.
2. We are using Alice's electron as the control qubit. So, we get $\alpha|00\rangle + \beta|11\rangle$. This is an entangled state because whenever Alice's measurement is 0 (or 1), Bob's is also 0 (or 1).

Pic.20 Quantum teleportation algorithm for transferring a quantum state from Alice's lab to Bob's lab

Quantum Teleportation

The algorithm for a simplified version of quantum teleportation:

- Step 2: Alice Measures Her Qubits in the $|+\rangle / |-\rangle$ Basis



Pic.20 Quantum teleportation algorithm for transferring a quantum state from Alice's lab to Bob's lab

After Alice has entangled her qubit with Bob's, Alice will try to measure her qubit in the $|+\rangle / |-\rangle$ basis. Since $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, we have

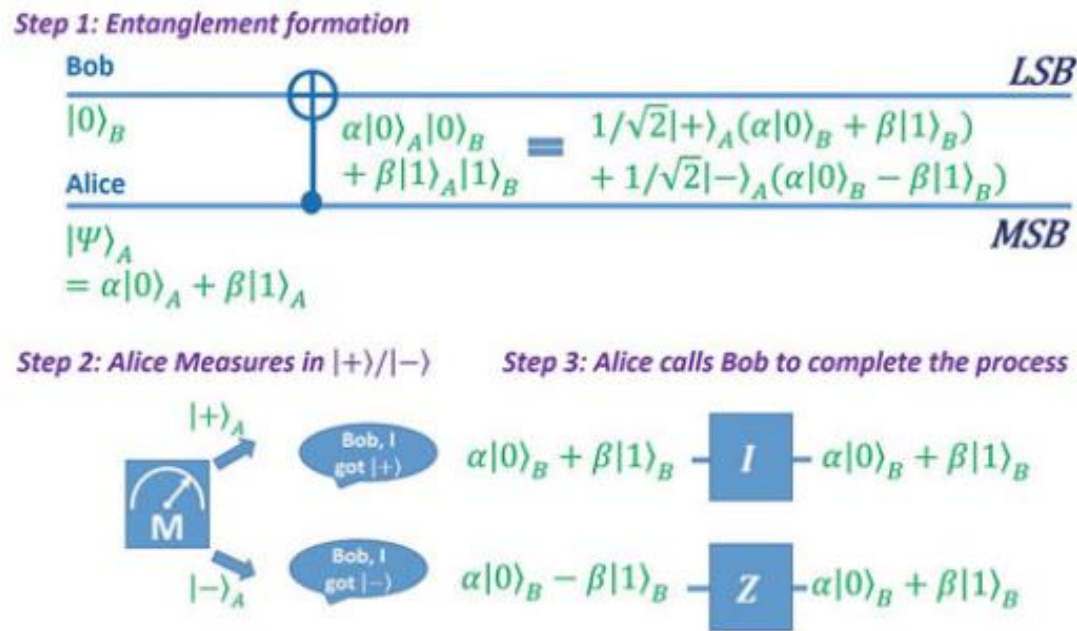
$$\begin{aligned} \alpha|00\rangle + \beta|11\rangle &= \alpha|0\rangle_A|0\rangle_B + \beta|1\rangle_A|1\rangle_B \\ &= \alpha \left(\frac{1}{\sqrt{2}}(|+\rangle_A + |-\rangle_A) \right) |0\rangle_B \\ &\quad + \beta \left(\frac{1}{\sqrt{2}}(|+\rangle_A - |-\rangle_A) \right) |1\rangle_B \\ &= |+\rangle_A \frac{1}{\sqrt{2}}(\alpha|0\rangle_B + \beta|1\rangle_B) \\ &\quad + |-\rangle_A \frac{1}{\sqrt{2}}(\alpha|0\rangle_B - \beta|1\rangle_B) \end{aligned}$$

Now if Alice performs a measurement on her qubit in the $|+\rangle / |-\rangle$ basis, she will obtain either $|+\rangle_A$ or $|-\rangle_A$ and Bob's qubit will become either $\alpha|0\rangle_B + \beta|1\rangle_B$ or $\alpha|0\rangle_B - \beta|1\rangle_B$.

Quantum Teleportation

The algorithm for a simplified version of quantum teleportation:

- Step 3: Classical Communication to Complete the Teleportation



Alice will then call Bob and let him know her outcome through a classical channel, which is slower than or the same as the speed of the light. If Alice obtains $|+\rangle$, Bob knows that his electron is in the state of $|\psi\rangle$, and the quantum teleportation process is completed. Or equivalently, he will apply an identity gate to his qubit. However, if Alice obtains $|-\rangle$, Bob knows his electron is in the state of $\frac{1}{\sqrt{2}}(\alpha|0\rangle_B - \beta|1\rangle_B)$, which is not $|\psi\rangle$. He will apply a Z-gate, which is a phase shift gate with angle being π .

Pic.20 Quantum teleportation algorithm for transferring a quantum state from Alice's lab to Bob's lab

Now, Alice's qubit's original state has also been destroyed due to the measurement.

Quantum Teleportation

- **Measurement in the $|+\rangle / |-\rangle$ Basis**

- In the quantum teleportation algorithm, Alice needs to measure her qubit in the $|+\rangle / |-\rangle$ basis in Step 2. If the apparatus can only measure in the $|0\rangle / |1\rangle$ basis, how do we achieve this?
- The trick is to apply the Hadamard gate before the measurement.
- In Step 2, instead of just performing a measurement in the $|+\rangle / |-\rangle$ basis, a Hadamard gate is applied followed by a measurement in the $|0\rangle / |1\rangle$ basis.
- For an arbitrary wavefunction $|\phi\rangle = a|0\rangle + b|1\rangle$, the probability distribution in the $|+\rangle / |-\rangle$ basis measurement is equivalent to that in the $|0\rangle / |1\rangle$ basis measurement after it is transformed by a Hadamard gate
- We first represent $|\phi\rangle$ in the $|+\rangle / |-\rangle$ basis to find out the probabilities of getting $|+\rangle$ and $|-\rangle$. Since, $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, we have

$$\begin{aligned}
 |\phi\rangle &= a|0\rangle + b|1\rangle \\
 &= a\left(\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)\right) + b\left(\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)\right) \\
 &= \frac{1}{\sqrt{2}}((a+b)|+\rangle + (a-b)|-\rangle)
 \end{aligned}$$

- If we perform a measurement on the $|+\rangle / |-\rangle$ basis, the probability of getting $|+\rangle$ is $|a+b|^2/2$ and that for $|-\rangle$ is $|a-b|^2/2$

Quantum Teleportation

- **Measurement in the $|+\rangle / |-\rangle$ Basis**

- What if we apply a Hadamard gate to $|\phi\rangle$ and then perform the measurement in the $|0\rangle / |1\rangle$ basis?

$$\begin{aligned} H|\phi\rangle &= a|+\rangle + b|-\rangle \\ &= a\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) + b\left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) \\ &= \frac{1}{\sqrt{2}}((a+b)|0\rangle + (a-b)|1\rangle) \end{aligned}$$

- If we perform a measurement on the $|0\rangle / |1\rangle$ basis, the probability of getting $|0\rangle$ is $|a+b|^2/2$ and that for $|1\rangle$ is $|a-b|^2/2$. This is the same as the distribution of performing a $|\phi\rangle$ measurement on the $|+\rangle / |-\rangle$ basis.
- Since this is a 2-qubit system, the basis states are either in the form of $|1X\rangle$ or $|0X\rangle$, where X can be 0 or 1. Since $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, if it is $|1X\rangle$, we can replace the first qubit (Alice's qubit) accordingly and get $\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) \otimes |X\rangle$. If we perform a measurement using the $|+\rangle / |-\rangle$ basis, it will collapse to either $|+\rangle|X\rangle$ or $-|-\rangle|X\rangle$. From Bob's perspective, his qubit becomes $|X\rangle$ or $-|X\rangle$, respectively.
- If we apply a Hadamard gate to Alice's qubit first, $|1X\rangle$ becomes $|-\rangle|X\rangle$, which is $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |X\rangle$. When Alice performs a measurement using the $|0\rangle / |1\rangle$ basis, it will collapse to either $|0\rangle|X\rangle$ or $-|1\rangle|X\rangle$. From Bob's perspective, his qubit again becomes $|X\rangle$ or $-|X\rangle$, respectively.
- Therefore, the effect on the LSB is the same. So, performing a $|+\rangle / |-\rangle$ measurement is equivalent to performing a $|0\rangle / |1\rangle$ measurement after the application of the Hadamard gate.

Chapter 19: Quantum Teleportation II and Entanglement Swapping

Kamonluk Suksen Ph.D.

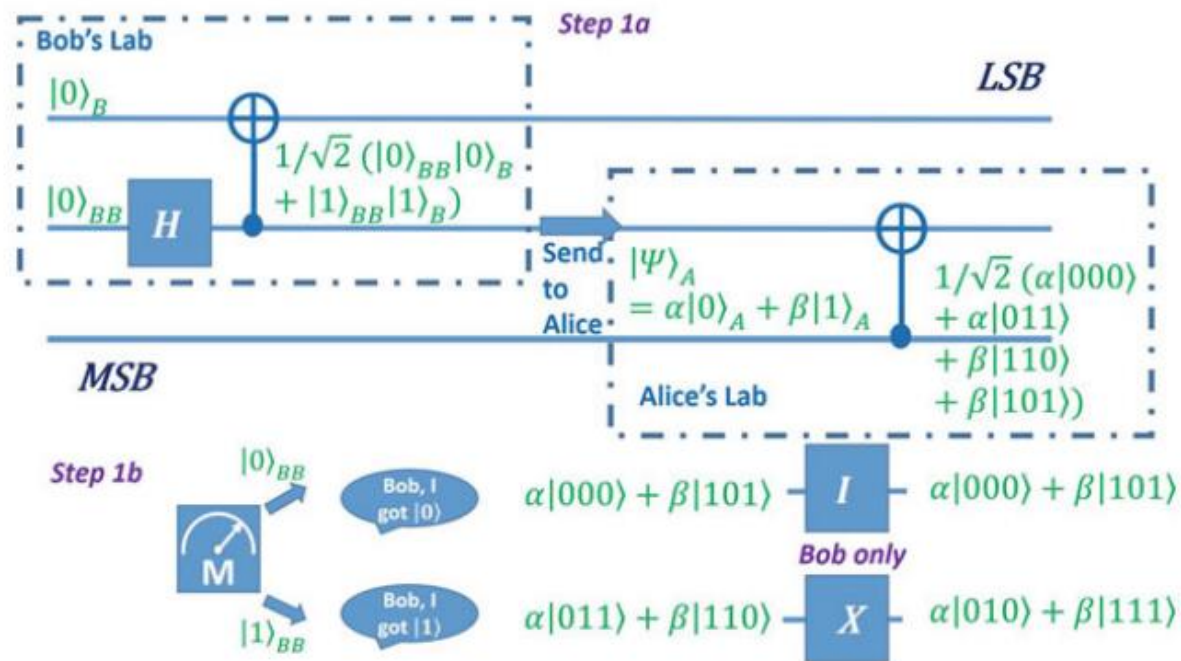
Quantum Teleportation: The Full Version

- In the full version of quantum teleportation, we will solve the difficulty in creating an ideal CNOT gate that needs to operate on two qubits separated by a huge distance.
- We will involve one more qubit (an ancillary qubit) and also need to physically transport that qubit from Bob's lab to Alice's lab before performing any quantum teleportation. This means we must prepare for teleportation ahead by transporting an ancillary qubit to the target lab.
- An ancillary qubit is an “extra” qubit but is critical for the circuit to achieve the goal. Therefore, although it is extra, it is not redundant.
- For example, in the quantum teleportation circuit, Alice's qubit is the source and Bob's qubit is the target. Therefore, these are the only two qubits we need for quantum teleportation. However, if we want to obviate the use of the ideal CNOT gate, we need to add an extra ancillary bit.
- In the complete algorithm, we will replace Step 1 with Step 1a and Step 1b. The purpose is to create the entanglement in Step 1 of the simplified version.

Quantum Teleportation

The algorithm for a full version of quantum teleportation:

- Step 1a: Qubit Entanglement and Ancillary Qubit Transportation



Bob will entangle the qubit $|0\rangle_B$, he wants to write ahead with another ancillary qubit initialized as $|0\rangle_{BB}$. The qubit he want to write is the LSB. Therefore, the state of Bob's qubits is $|0\rangle_{BB} \otimes |0\rangle_B$. He then applies $H \otimes I$ to it. Therefore, it becomes $\frac{1}{\sqrt{2}} (|0\rangle_{BB} + |1\rangle_{BB}) \otimes |0\rangle_B$. So it can be written as $\frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$. Then it goes through the CNOT gate. Note that both qubits are in close proximity and they are both in Bob's lab. So there is no problem in creating a CNOT gate with high fidelity. So, the state of Bob's qubits is now $\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$, which is entangled ($|\phi^+\rangle$).

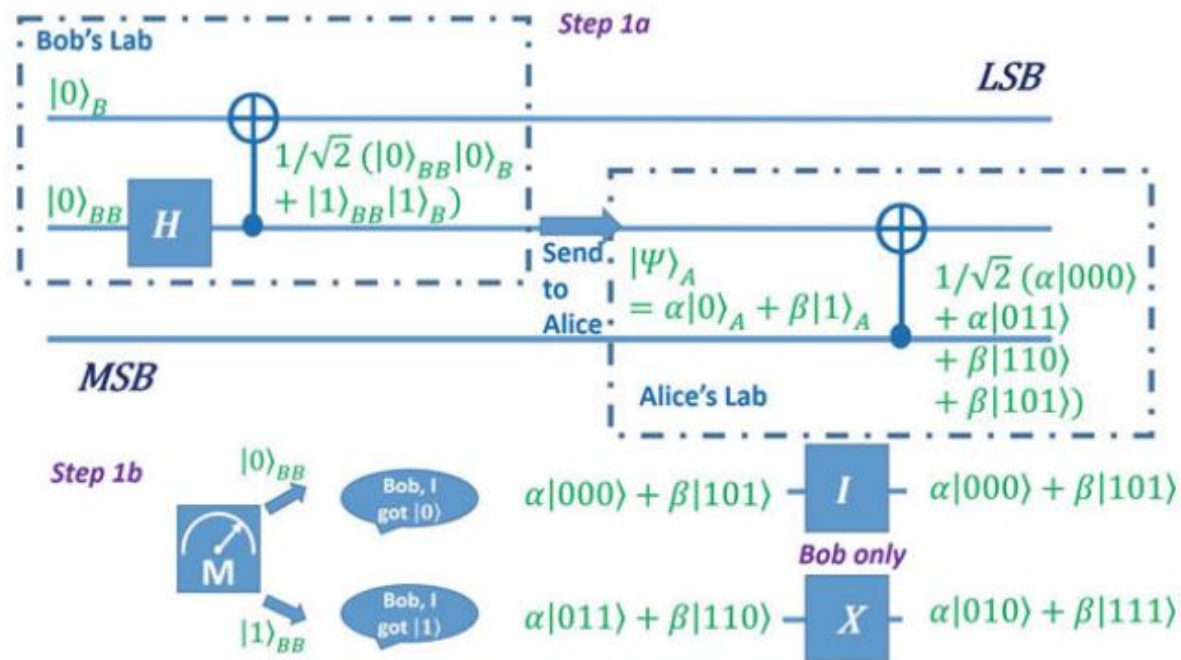
Pic.21 Algorithm to replace Step 1 in the simplified quantum teleportation circuit in Pic. 20.

After this, Steps 2 and 3 in Pic. 20 need to be executed to complete the quantum teleportation process

Quantum Teleportation

The algorithm for a full version of quantum teleportation:

- Step 1a: Qubit Entanglement and Ancillary Qubit Transportation



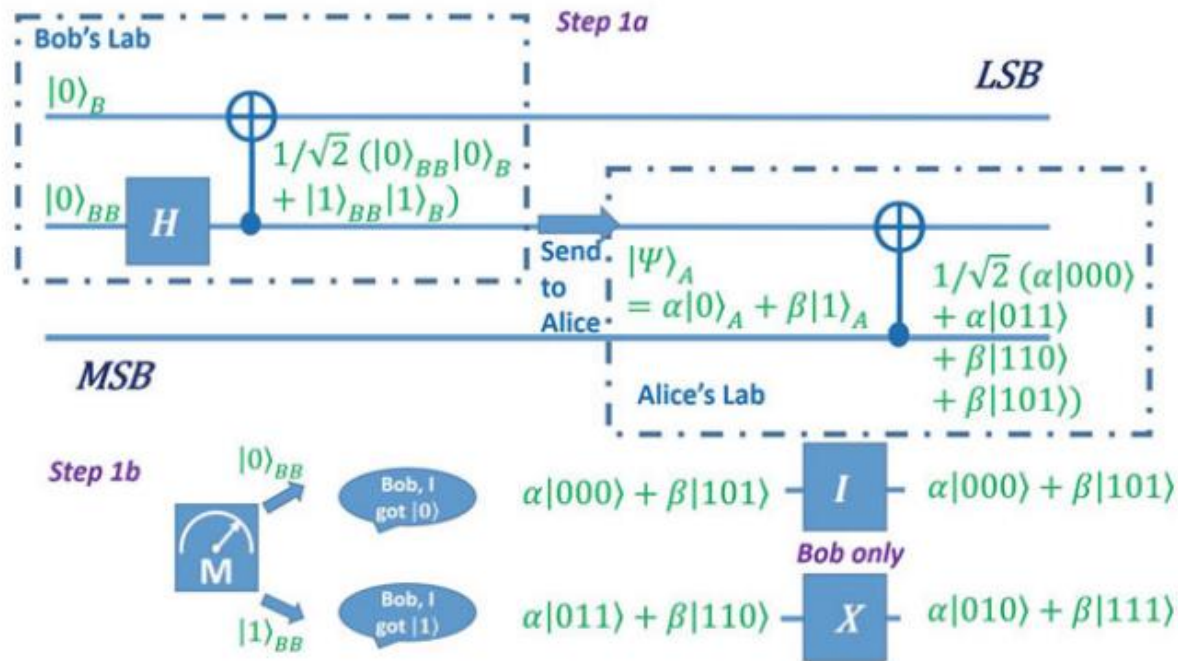
Now Bob will send the BB qubit to Alice and it will stay there and still entangled with the B qubit in Bob's lab until they want to do quantum teleportation. One day, Alice decides to quantum teleport her qubit information to Bob. That qubit has a state $|\psi\rangle_A = \alpha|0\rangle_A + \beta|1\rangle_A$ like in the simplified version. She will then perform a CNOT operation, by using her qubit A as the control qubit, on qubit BB which was sent by Bob earlier and is still entangled with Bob's qubit B .

Pic.21 Algorithm to replace Step 1 in the simplified quantum teleportation circuit in Pic. 20. After this, Steps 2 and 3 in Pic. 20 need to be executed to complete the quantum teleportation process

Quantum Teleportation

The algorithm for a full version of quantum teleportation:

- Step 1a: Qubit Entanglement and Ancillary Qubit Transportation



Pic.21 Algorithm to replace Step 1 in the simplified quantum teleportation circuit in Pic. 20. After this, Steps 2 and 3 in Pic. 20 need to be executed to complete the quantum teleportation process

Before the CNOT gate by Alice, the state of the 3-qubit system is:

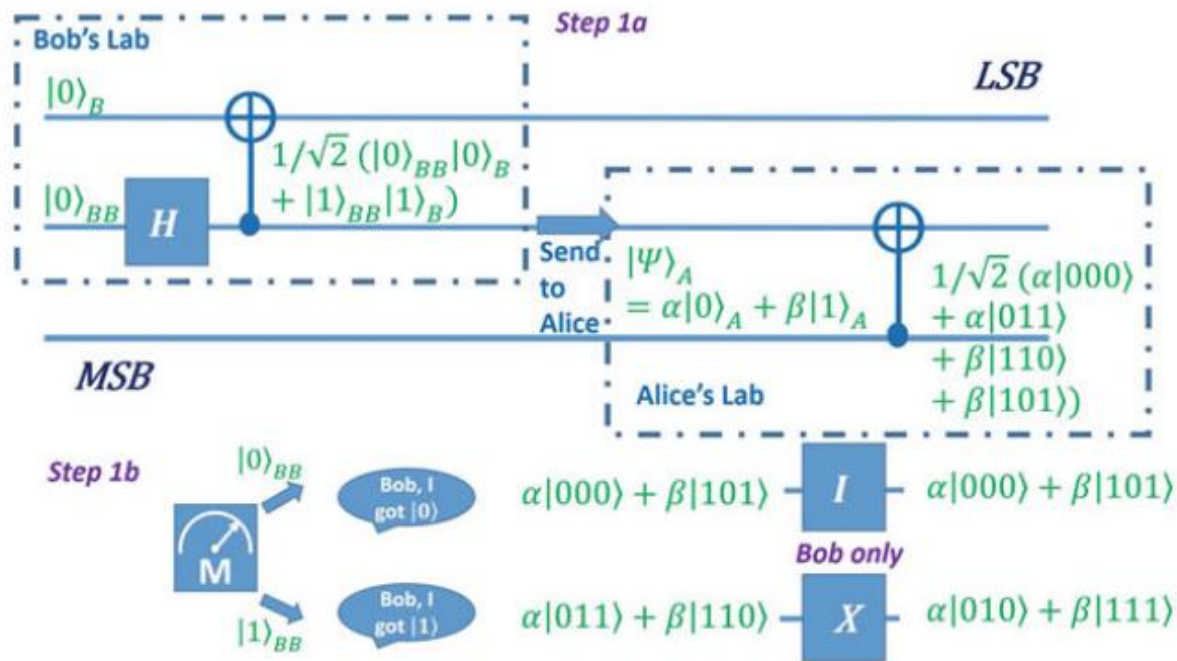
$$\begin{aligned}
 & |\psi\rangle_A \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\
 &= (\alpha|0\rangle_A + \beta|1\rangle_A) \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\
 &= (\alpha|0\rangle + \beta|1\rangle) \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\
 &= \frac{1}{\sqrt{2}} (\alpha|000\rangle + \beta|100\rangle + \alpha|011\rangle + \beta|111\rangle)
 \end{aligned}$$

After that, the CNOT gate is applied. The control qubit is the MSB (A) and the second qubit (BB) is the target qubit. Whenever the basis MSB is 1, the target qubit will be flipped. So, $|100\rangle$ becomes $|110\rangle$ and $|111\rangle$ becomes $|101\rangle$. As a result, the state of the 3-qubit system becomes, $\frac{1}{\sqrt{2}} (\alpha|000\rangle + \beta|101\rangle + \alpha|011\rangle + \beta|110\rangle)$.

Quantum Teleportation

The algorithm for a full version of quantum teleportation:

- Step 1b: Measurement of the Ancillary Qubit



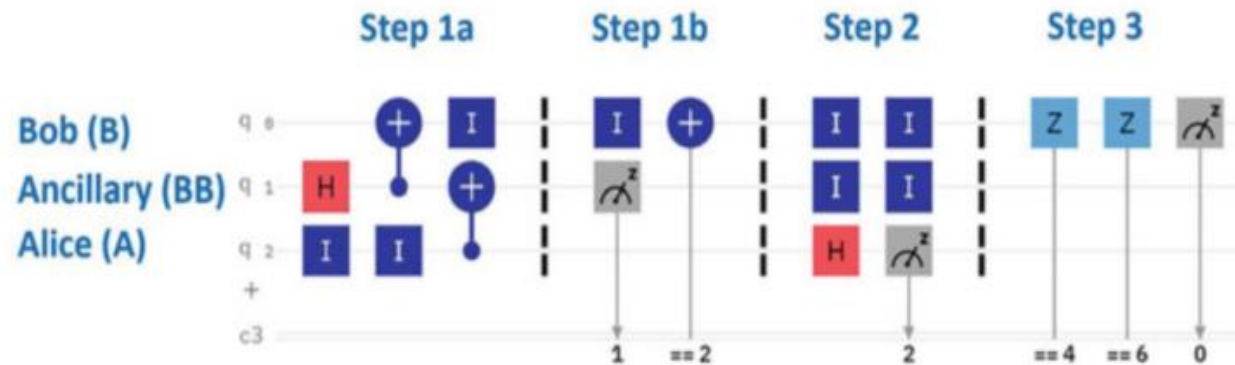
Pic.21 Algorithm to replace Step 1 in the simplified quantum teleportation circuit in Pic. 20. After this, Steps 2 and 3 in Pic. 20 need to be executed to complete the quantum teleportation process

Now Alice will perform a measurement on the second qubit (BB) on the $|0\rangle / |1\rangle$ basis. If she obtains $|0\rangle$, it means the whole system has collapsed to a state containing only $|000\rangle$ and $|101\rangle$, which is $\alpha|000\rangle + \beta|101\rangle$ (after normalization).

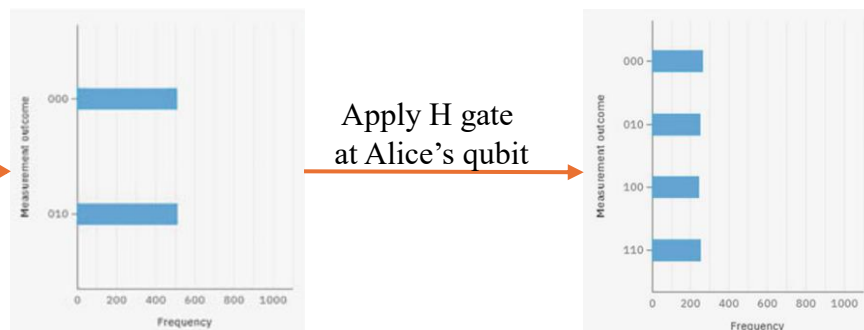
However, Alice can just call Bob and tell him that she got $|0\rangle$. Their qubits have been entangled and they can continue to perform Step 2 and Step 3 in Pic. 20 to complete the quantum teleportation. If Alice, on the other hand, obtains $|1\rangle$, Bob just needs to apply a NOT gate to his qubit to flip the bit value in the basis states. If only consider the MSB and LSB, they are $\alpha|01\rangle + \beta|10\rangle$. They are still entangled because when the MSB is 1 (or 0), the LSB is always 0 (or 1). Therefore, the state becomes $\alpha|00\rangle + \beta|11\rangle$, which is the output of Step1 in the simplified version in Pic. 20.

Quantum Teleportation

The algorithm for a full version of quantum teleportation:



Pic.22 Implementation of the full quantum teleportation algorithm by combining Pic. 21 and Steps 2 and 3 in Pic. 20



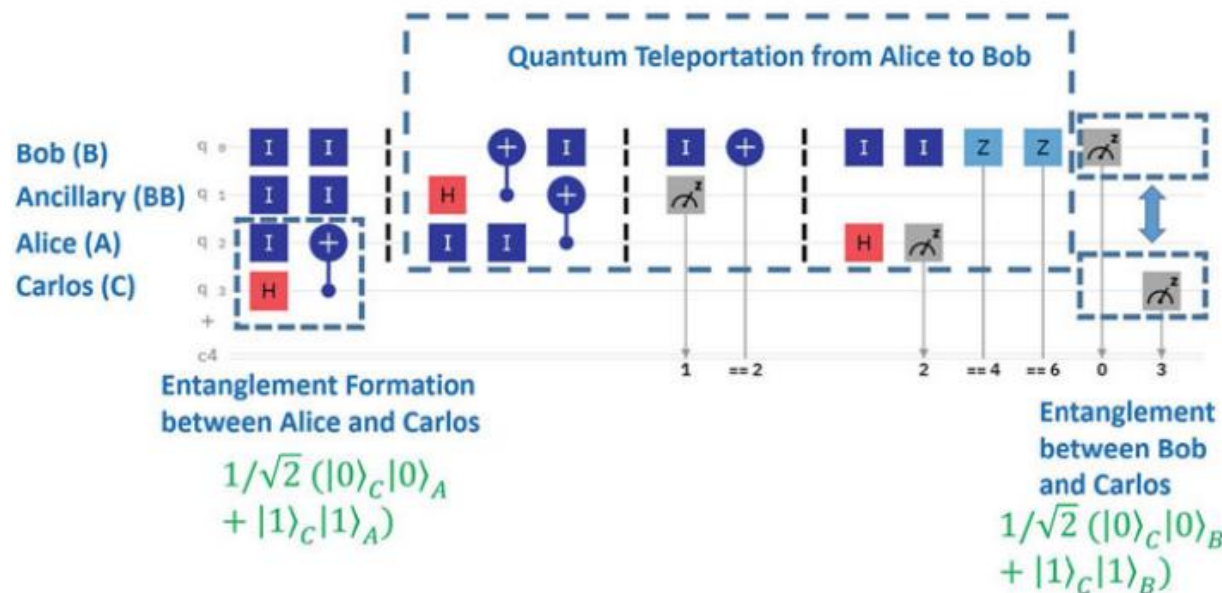
Pic. 22 shows the quantum circuit built-on IBM-Q. It has four steps. Step 1a and Step 1b correspond to Pic. 21 to perform entanglement between Alice's qubit (A) and Bob's qubit (B) with the help of the ancillary qubit (BB) to obviate the need of an ideal CNOT gate.

Step 2 and Step 3 implement the Step 2 and Step 3 in Pic. 20. It is worth mentioning that the ancillary qubit goes through only identity gates and has no contribution to the algorithm anymore after Step 1b.

In Step 1b, Alice measures q1 using the $|0\rangle/|1\rangle$ basis. As the setup shows, it is stored in 1st classical bit. Therefore, if she measures $|1\rangle$, c3 becomes $(010)_2$. If she measures $|0\rangle$, c3 is still $(000)_2$. Then Bob will apply a NOT-gate if the value in c3 is 2 (in decimal), which is $(010)_2$ and doing nothing otherwise.

In Step 2, Alice performs a measurement on the source qubit (A) using the $|+\rangle/|-\rangle$ basis. The Hadamard gate is used so that we can perform a $|+\rangle/|-\rangle$ basis measurement. The measurement result is stored in the 2nd bit in the classical register (MSB). Bob will apply Z-gate if Alice measures $|1\rangle$ in Step 2. This means c3 can be either $(100)_2$ or $(110)_2$, which are 4 or 6 in decimal, respectively. Therefore, in Step 3, two conditional Z gates are used and they are only applied when either $c3 = 4$ or $c3 = 6$. Note that for every single experiment, c3 cannot be 4 and 6 at the same time. Therefore, the Z-gate will never be applied twice.

Entanglement Swapping

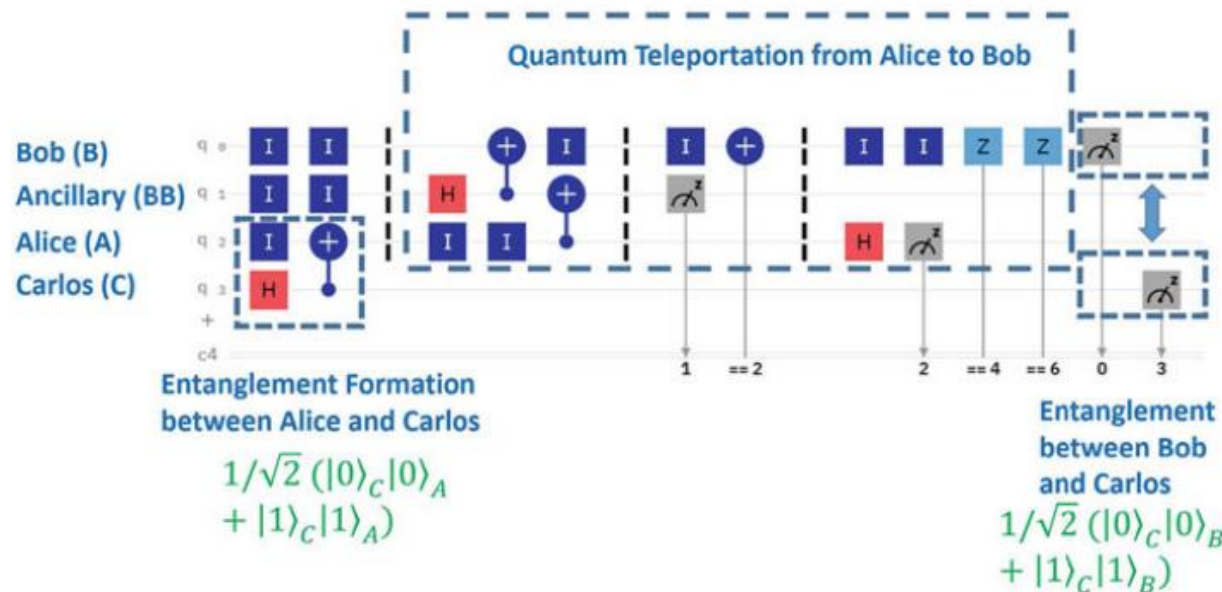


Pic.23 Entanglement swapping circuit built in IBM-Q

Assume Alice's electron was entangled with another electron, owned by Carlos (our new character in this play). If Alice teleports the state of her electron to Bob, we will expect that Bob's electron will then entangle with Carlos' electron. This is a natural consequence if we agree quantum teleportation of a state in the microscopic world is very similar to hypothetical teleportation. Now we also have Carlos's electron and this becomes a 4-qubit system.

If we teleport Alice's qubit to Bob while Alice's qubit was entangled with Carlos' qubit, Bob's qubit will then entangle with Carlos' qubit after quantum teleportation.

Entanglement Swapping



Pic.23 Entanglement swapping circuit built in IBM-Q

Use matrix multiplication to show that Alice's and Carlos' are entangled right before the quantum teleportation block in Pic.22.

The input state to the circuit is $|0\rangle_C \otimes |0\rangle_A \otimes |0\rangle_{BB} \otimes |0\rangle_B$. The part of the circuit performing the operation is $(U_{XOR} \otimes I^{\otimes 2})(H \otimes I^{\otimes 3})$.

This is a 4-qubit system and the matrix will be 16×16 . We see that only identity gates are applied to the last two qubits (the ancillary bit BB and Bob's B), therefore,

$$\begin{aligned}
 & (U_{XOR} \otimes I^{\otimes 2})(H \otimes I^{\otimes 3})|0000\rangle \\
 &= ((U_{XOR})(H \otimes I)|00\rangle) \otimes (I^{\otimes 2}I^{\otimes 2}|00\rangle) \\
 &= ((U_{XOR})(H \otimes I)|00\rangle) \otimes (|00\rangle) \\
 &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \otimes (|00\rangle)
 \end{aligned}$$

So, Carlos' and Alice's qubits are entangled as whenever Carlos measures $|0\rangle$ (or $|1\rangle$), Alice's qubit will collapse to $|0\rangle$ (or $|1\rangle$). Now including the BB and C qubits, the state is $\frac{1}{\sqrt{2}}(|0000\rangle + |1100\rangle)$

Chapter 20:

Deutsch Algorithm

Kamonluk Suksen Ph.D.

Deutsch Algorithm

- The Deutsch Algorithm is a sub-problem of a more general Deutsch–Jozsa Algorithm. We will only discuss the Deutsch Algorithm and inspect the algorithm in detail.
- The Deutsch Algorithm provides a two-time speedup in the computation.
- In the more general Deutsch–Jozsa Algorithm, it provides an exponential speedup over the classical one. This means that for an n -qubit problem, classically we need $2^{n-1} + 1$ computations but only 1 computation using the Deutsch–Jozsa Algorithm.
- Therefore, the Deutsch Algorithm also provides an exponential speedup, but it just happens that $n = 1$ and thus it only has two times of speedup.
- In the Deutsch–Jozsa problem, we are given an oracle (a black box into which we cannot peep), which maps $f: \{0, 1\}^n \rightarrow \{0, 1\}$. it takes an n binary digits (n -bits) as the input.
- For example, if $n = 3$, the inputs can be any permutation of 3 digits of 0 or 1. E.g. the input may be “000”, “001”, ..., “111”.
- So, an oracle is just a *function* and we do not know which function it implements. But we are told that it is either balanced or constant in the Deutsch–Jozsa problem (this is given).
- A balanced function means that half of its outputs are “0” and the other half are “1.”
- A constant function means that, regardless of the input values, it always outputs the same value (a constant value). The output is constantly “1” or constantly “0”.
- *The Deutsch–Jozsa Algorithm is to figure out if the oracle is balanced or constant.* The problem is completely classical.

Deutsch Algorithm

- Give an example of an oracle that is constant for the $n = 3$ case.
- There are eight possible inputs in the $n = 3$ case.
- Let the input bits be x_2 , x_1 , and x_0 .
- If an oracle implements the following function: $f(x_2 x_1 x_0) = x_2 \cdot x_1 \cdot x_0 \cdot 0$, where “.” is the classical Boolean AND operation, then

$$f(000) = 0.0.0.0 = 0$$

$$f(001) = 0.0.1.0 = 0$$

$$f(010) = 0.1.0.0 = 0$$

$$f(011) = 0.1.1.0 = 0$$

$$f(100) = 1.0.0.0 = 0$$

$$f(101) = 1.0.1.0 = 0$$

$$f(110) = 1.1.0.0 = 0$$

$$f(111) = 1.1.1.0 = 0$$

And this is a *constant* oracle, which always gives “0” as the output.

Deutsch Algorithm

- Give an example of an oracle that is constant for the $n = 3$ case.
- There are eight possible inputs in the $n = 3$ case.
- Let the input bits be x_2 , x_1 , and x_0 .
- If an oracle implements the following function: $f(x_2 x_1 x_0) = x_2 \oplus x_1 \oplus x_0$, where “ $\oplus$ ” is the classical XOR operation, then

$$f(000) = 0 \oplus 0 \oplus 0 = 0$$

$$f(001) = 0 \oplus 0 \oplus 1 = 1$$

$$f(010) = 0 \oplus 1 \oplus 0 = 1$$

$$f(011) = 0 \oplus 1 \oplus 1 = 0$$

$$f(100) = 1 \oplus 0 \oplus 0 = 1$$

$$f(101) = 1 \oplus 0 \oplus 1 = 0$$

$$f(110) = 1 \oplus 1 \oplus 0 = 0$$

$$f(111) = 1 \oplus 1 \oplus 1 = 1$$

This is a *balanced* oracle. A half (4) of the inputs results in “0” and a half (4) of the inputs results in “1”.

Deutsch Algorithm

- We do NOT know what function the oracle implements. But we are told that it can be either balanced or constant.
- Now, let us turn to the Deutsch problem. It is a special case of the Deutsch–Jozsa problem with $n = 1$.
- Therefore, the oracle maps $f: \{0, 1\} \rightarrow \{0, 1\}$. It only has 1 bit of input and thus it only has two possible inputs (either “0” or “1”).
- As a result, it only has two possible outputs ($f(0)$ and $f(1)$). Our goal is still to find out if the oracle is “*balanced*” or “*constant*” ($f(0) = f(1)$).
- Since it only has two possible outputs, the “balanced” case is the same as a “non-constant” case as there is only one non-constant case, i.e. ($f(0) \neq f(1)$).
- While in the Deutsch–Jozsa problem with $n > 1$, non-constant is not necessarily balanced.

Deutsch Algorithm

- **Classical Algorithm**

- To find out if the solution is constant, in a classical way, we need to provide the input one by one and measure the outputs.
- If both outputs are the same, then it is a constant oracle. Otherwise, it is balanced (or non-constant).
- Therefore, we need to use the oracle to perform two computations to know the answer.
- For example, in the Deutsch problem, what are the possible input and output combinations?
Since there are two inputs and two outputs, there are only 4 types of functions.

$$f_A(0) = 0 \text{ and } f_A(1) = 0$$

$$f_B(0) = 0 \text{ and } f_B(1) = 1$$

$$f_C(0) = 1 \text{ and } f_C(1) = 0$$

$$f_D(0) = 1 \text{ and } f_D(1) = 1$$

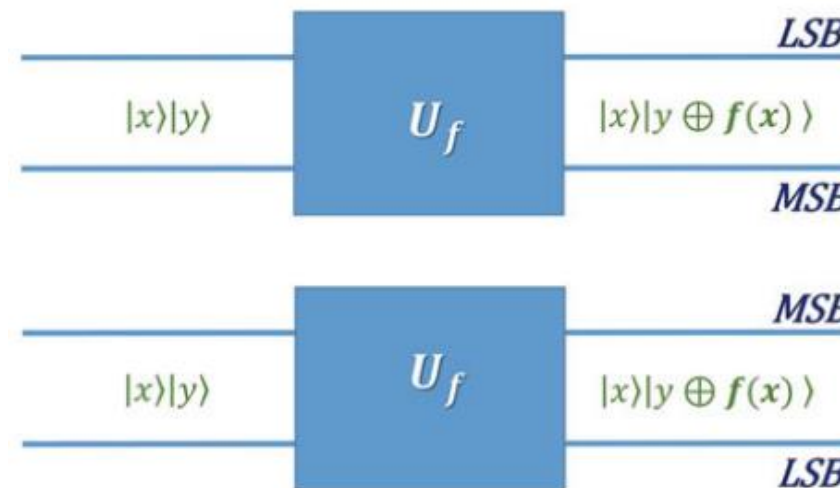
We see that f_A and f_D are constant functions while f_B and f_C are balanced functions (or non-constant functions).

- There is no way that we can determine if the function is constant or not without performing two calculations (i.e. computing $f(0)$ and $f(1)$).

Deutsch Algorithm

• Quantum Computing Solution

- Luckily, we have quantum computers. By using the Deutsch Algorithm, we only need to perform one computation by using **quantum parallelism**.
- **Step 0: Create a Quantum Oracle**
We will discuss the **quantum oracle** in detail in the next chapter. Now we will only learn the essence of quantum oracle so we can understand the Deutsch algorithm.
- Pic. 24 shows a quantum oracle embedded in a quantum circuit of 2 qubits. A quantum oracle is just a quantum gate! It needs to be unitary. The quantum oracle encodes the problem we are trying to solve.



Pic.24 The quantum oracle of a 2-qubit circuit used in the Deutsch algorithm. The top is the convention we have been using (IBM-Q) with MSB at the bottom and the bottom one is the other version often seen in other books. Both are using the little-endian convention. Note that $|x\rangle|y\rangle$ are basis vectors

Deutsch Algorithm

• Quantum Computing Solution

▪ Step 0: Create a Quantum Oracle

Therefore, we will just treat it as a black box and we assume some physical systems in nature can be used to create this oracle in the quantum computer if we do not know how to create it from simple quantum gates.

- Since a quantum oracle is just a quantum gate, the definition of a quantum oracle is still defined by how it transforms the basis states just like how we defined a simple quantum gate. Pic. 24 shows the definition.
- The symbol of the oracle is U_f , signifying that this is a unitary gate corresponding to the oracle function, $f(x)$.
- For any **basis vector**, $|x\rangle|y\rangle$, the output is $|x\rangle|y \oplus f(x)\rangle$, where $\oplus$ is the classical XOR and $f(x)$ is the function, we are evaluating. That is

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

- This means that when the input is a basis vector with x being the MSB and y being the LSB, the output will be another basis vector with the MSB unchanged (still x) but the LSB becomes $y \oplus f(x)$, which depends on the $f(x)$ of the oracle.
- For example, if $f(x)$ is a constant function that always gives “0”, how does the oracle transform the basis vectors? There are only four basis vectors in a 2-qubit system, namely $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. By tracing the change of x and y :

$$U_f|0\rangle|0\rangle = |0\rangle|0 \oplus f(0)\rangle = |0\rangle|0 \oplus 0\rangle = |0\rangle|0\rangle$$

$$U_f|0\rangle|1\rangle = |0\rangle|1 \oplus f(0)\rangle = |0\rangle|1 \oplus 0\rangle = |0\rangle|1\rangle$$

$$U_f|1\rangle|0\rangle = |1\rangle|0 \oplus f(1)\rangle = |1\rangle|0 \oplus 0\rangle = |1\rangle|0\rangle$$

$$U_f|1\rangle|1\rangle = |1\rangle|1 \oplus f(1)\rangle = |1\rangle|1 \oplus 0\rangle = |1\rangle|1\rangle$$

We see that we can implement this oracle by an identity gate. But we still do not know what is $f(x)$ in the oracle. We only know how it behaves.

Deutsch Algorithm

• Quantum Computing Solution

▪ Step 1: Create a Superposition Input

We will create a special superposition state as the input to the oracle. It is not surprising that a superposition state is needed because we want to use quantum parallelism to evaluate the function with different basis vectors at the same time.

- The superposition state to be created is $|+\rangle|-\rangle$. We represent it in the original $|0\rangle/|1\rangle$ basis as:

$$\begin{aligned} |+\rangle|-\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) \end{aligned}$$

We can see that it has equal weight for every basis vector, but the phases are not the same.

▪ Step 2: Evaluate Using the Quantum Oracle

When apply U_f to $|+\rangle|-\rangle$:

$$\begin{aligned} U_f|+\rangle|-\rangle &= U_f\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)\right) \\ &= \frac{1}{2}(U_f|00\rangle - U_f|01\rangle + U_f|10\rangle - U_f|11\rangle) \\ &= \frac{1}{2}(|0, 0 \oplus f(0)\rangle - |0, 1 \oplus f(0)\rangle + |1, 0 \oplus f(1)\rangle - |1, 1 \oplus f(1)\rangle) \end{aligned}$$

- We know anything XOR with “0” is itself and anything XOR with “1” is its negation. Where $\overline{f(x)}$ is the negation of $f(x)$.

$$U_f|+\rangle|-\rangle = \frac{1}{2}(|0, f(0)\rangle - |0, \overline{f(0)}\rangle + |1, f(1)\rangle - |1, \overline{f(1)}\rangle)$$

Deutsch Algorithm

• Quantum Computing Solution

▪ Step 3: Measure on the Right Basis to Get the Answer

If $f(x)$ is a constant function, then $f(0) = f(1)$ and $\overline{f(0)} = \overline{f(1)}$. We can then perform a factorization on the tensor product:

$$\begin{aligned}
 U_f|+\rangle|-\rangle &= \frac{1}{2}(|0, f(0)\rangle - |0, \overline{f(0)}\rangle + |1, f(1)\rangle - |1, \overline{f(1)}\rangle) \\
 &= \frac{1}{2}(|0, f(0)\rangle - |0, \overline{f(0)}\rangle + |1, f(0)\rangle - |1, \overline{f(0)}\rangle) \\
 &= \frac{1}{2}(|0\rangle|f(0)\rangle - |0\rangle|\overline{f(0)}\rangle + |1\rangle|f(0)\rangle - |1\rangle|\overline{f(0)}\rangle) \\
 &= \frac{1}{2}((|0\rangle + |1\rangle)|f(0)\rangle - (|0\rangle + |1\rangle)|\overline{f(0)}\rangle) \\
 &= \frac{1}{\sqrt{2}}(|+\rangle|f(0)\rangle - |+\rangle|\overline{f(0)}\rangle) \\
 &= |+\rangle \frac{1}{\sqrt{2}}(|f(0)\rangle - |\overline{f(0)}\rangle)
 \end{aligned}$$

We also see that the oracle keeps the MSB unchanged if it is constant.

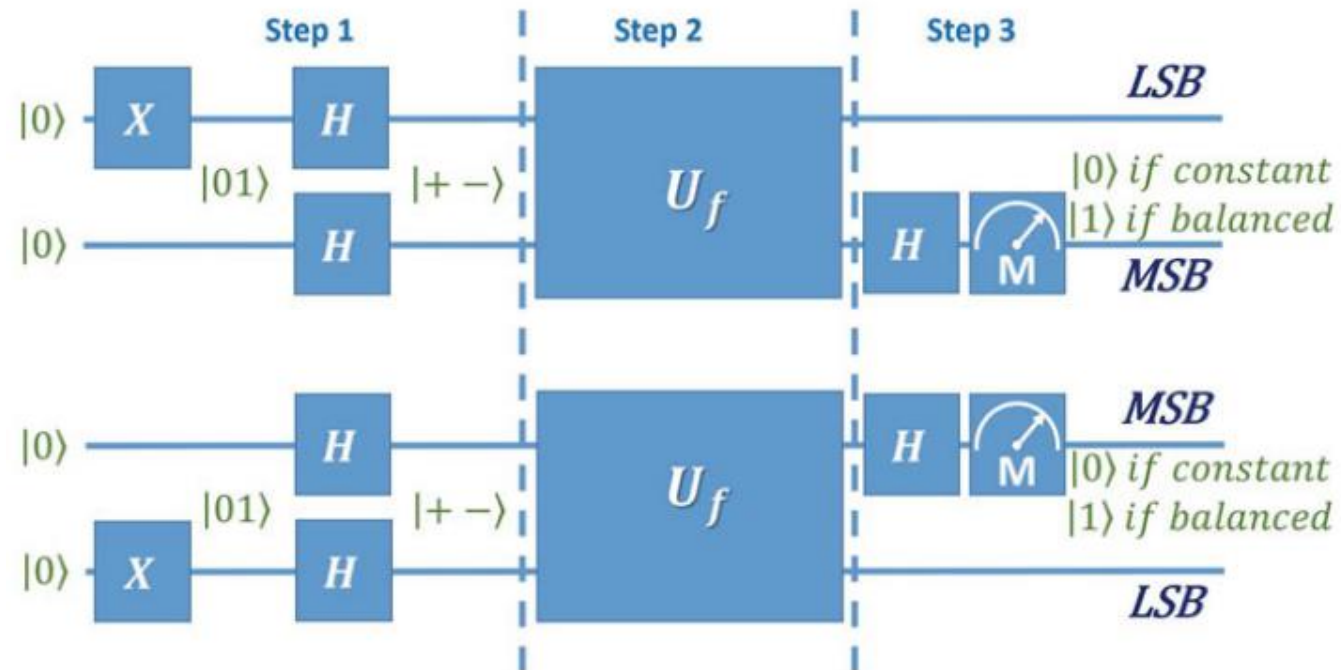
- If $f(x)$ is balanced (non-constant), then $f(0) = \overline{f(1)}$ and $\overline{f(0)} = f(1)$. For example, if $f(0) = 1$ and $f(1) = 0$, then $f(0) = \overline{f(1)} = \overline{0} = 1$. We can then perform a factorization on the tensor product:

$$\begin{aligned}
 U_f|+\rangle|-\rangle &= \frac{1}{2}(|0, f(0)\rangle - |0, \overline{f(0)}\rangle + |1, f(1)\rangle - |1, \overline{f(1)}\rangle) \\
 &= \frac{1}{2}((|0\rangle - |1\rangle)|f(0)\rangle - (|0\rangle - |1\rangle)|\overline{f(0)}\rangle) \\
 &= \frac{1}{\sqrt{2}}(|-\rangle|f(0)\rangle - |-\rangle|\overline{f(0)}\rangle) \\
 &= |-\rangle \frac{1}{\sqrt{2}}(|f(0)\rangle - |\overline{f(0)}\rangle)
 \end{aligned}$$

Deutsch Algorithm

• The Quantum Circuit

- In summary, the quantum oracle in the Deutsch algorithm takes $|+ -\rangle$ as the input and keeps the MSB as $|+\rangle$ if $f(x)$ is constant and flips it to $|-\rangle$ if it is balanced (non-constant). This is simple but we need to do two things to complete the circuit.

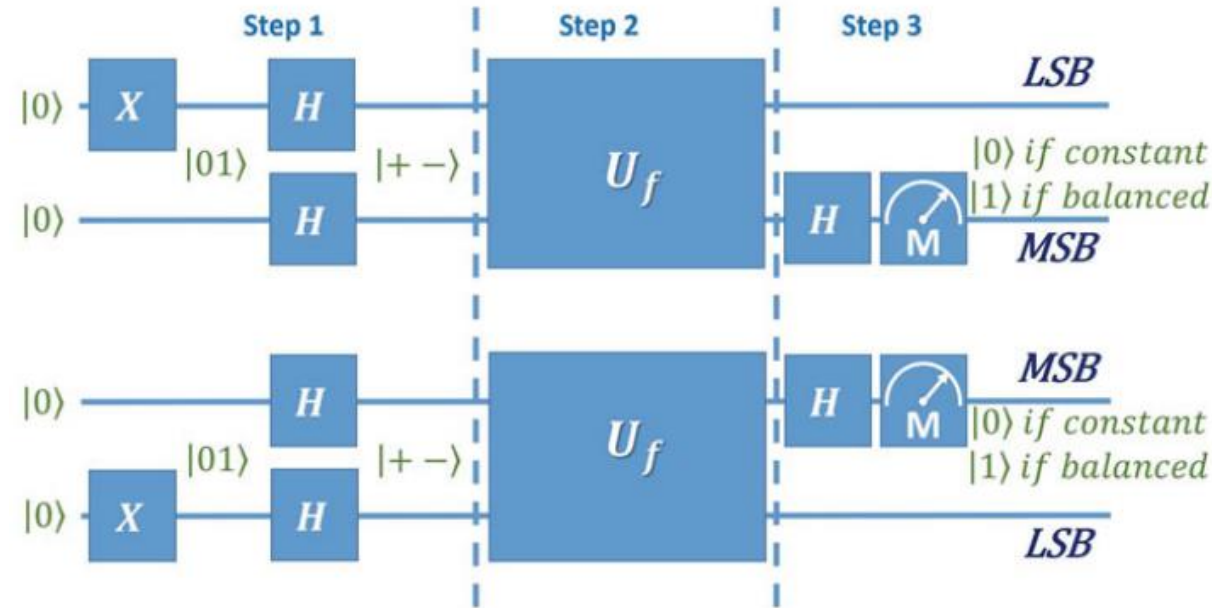


Pic. 25 The Deutsch algorithm circuit. The top is the convention we have been using (IBM-Q) with MSB at the bottom and the bottom one is the other version often seen in other books. Both are using little-endian convention

- Firstly, we need to create $|+ -\rangle$ and secondly, we need to measure in the $|0\rangle / |1\rangle$ basis to determine if the MSB is $|+\rangle$ or $|-\rangle$.

Deutsch Algorithm

- The Quantum Circuit



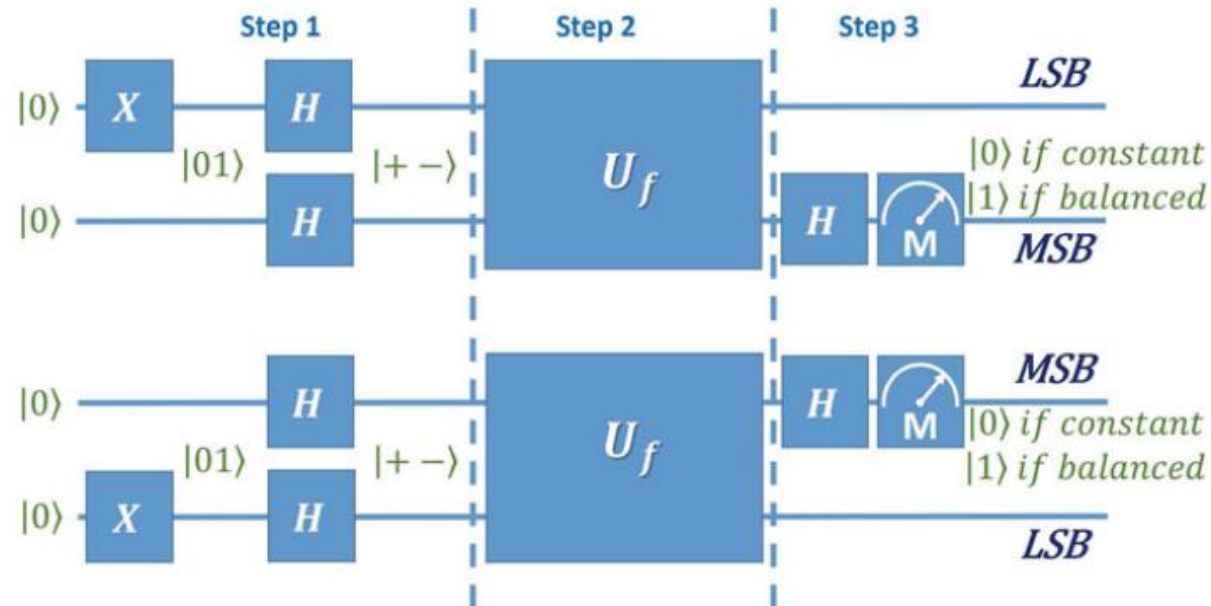
Pic. 25 The Deutsch algorithm circuit. The top is the convention we have been using (IBM-Q) with MSB at the bottom and the bottom one is the other version often seen in other books.

Both are using little-endian convention

- We prefer to start the quantum circuits by initializing the input qubits to $|0\rangle_n$ with $n = 2$ in this case, i.e. $|0\rangle|0\rangle$. We use the Hadamard gates to prepare $|+\rangle|-\rangle$ state. Since $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$. So, we need to create $|1\rangle$ first, using the NOT-gate (X).
- Step 1 of the circuit using the following equation: $(HH)(IX)|0\rangle|0\rangle = (HH)|0\rangle|1\rangle = |+\rangle|-\rangle$.

Deutsch Algorithm

- The Quantum Circuit



Pic. 25 The Deutsch algorithm circuit. The top is the convention we have been using (IBM-Q) with MSB at the bottom and the bottom one is the other version often seen in other books. Both are using little-endian convention

- How do we determine if the MSB is $|+\rangle$ or $|-\rangle$ while we can only measure in the $|0\rangle / |1\rangle$ basis? The trick is again to use the Hadamard gate. We discussed this in the quantum teleportation algorithm already. That is measuring a state in the $|+\rangle / |-\rangle$ basis is the same as measuring in the $|0\rangle / |1\rangle$ basis after applying a Hadamard gate. So, we apply a Hadamard gate after the quantum oracle and then perform a measurement on the MSB.
- If the result is $|0\rangle$ (corresponds to measuring $|+\rangle$ before the Hadamard gate), we know the function is constant. If the result is $|1\rangle$ (corresponds to measuring $|-\rangle$ before the Hadamard gate), we know the function is balanced.